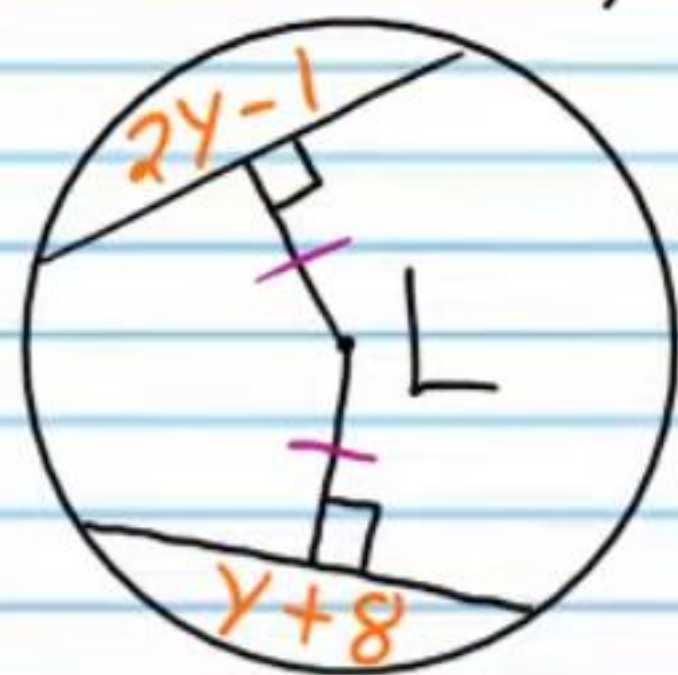


Practice Geometry Test V (Version I)

Name
Date
Course

- I) Write your name, the date, and the course title (i.e. Geometry).
- II) Show work when necessary.
- III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- IV) You will need a calculator for some problems on this test. When pi is required, use the π button on your calculator.
- V) Graphing paper may be useful for this test.
- VI) Each problem is worth four points for a total of 100 points.

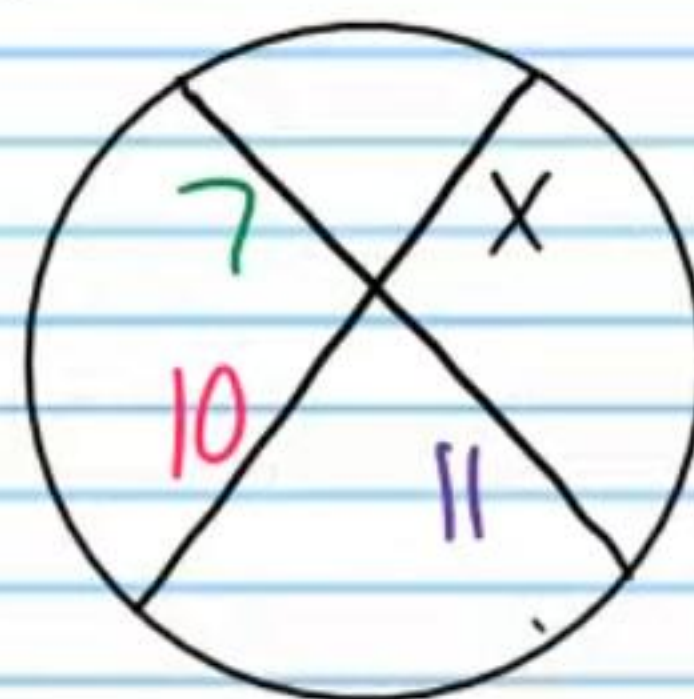
① If L is the center of the circle below, find y.



$$2y-1 = y+8$$

$$\boxed{y = 9}$$

② Find x.



$$7(11) = 10x$$

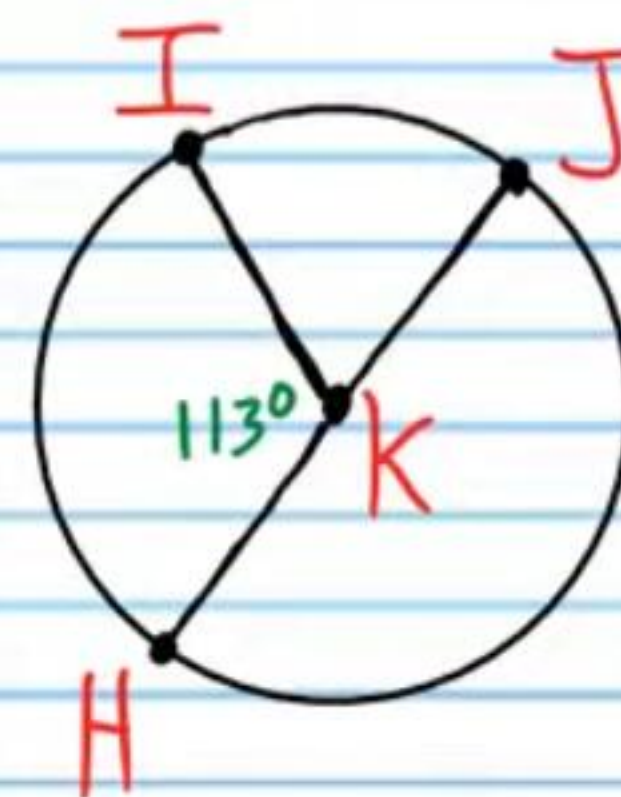
$$77 = 10x$$

$$\boxed{7.7 = x}$$

If K is the centers of the following circles, find the values below. If a problem requires the pi value, use the π button on your calculator. Round your answers to the nearest tenth if necessary.

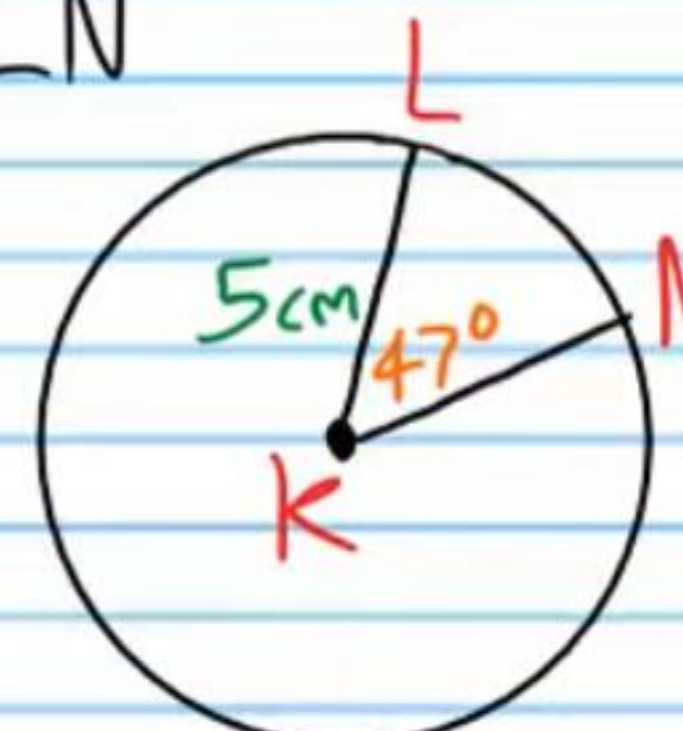
③ i) $m\widehat{HI}$ ii) $m\angle IKJ$ ④ $\mathcal{L}\widehat{LN}$

* $\overline{HJ}$ is a diameter.



$$\text{i) } m\widehat{HI} = \boxed{113^\circ}$$

$$\begin{aligned} \text{ii) } m\angle IKJ &= 180 - 113 \\ &= \boxed{67^\circ} \end{aligned}$$

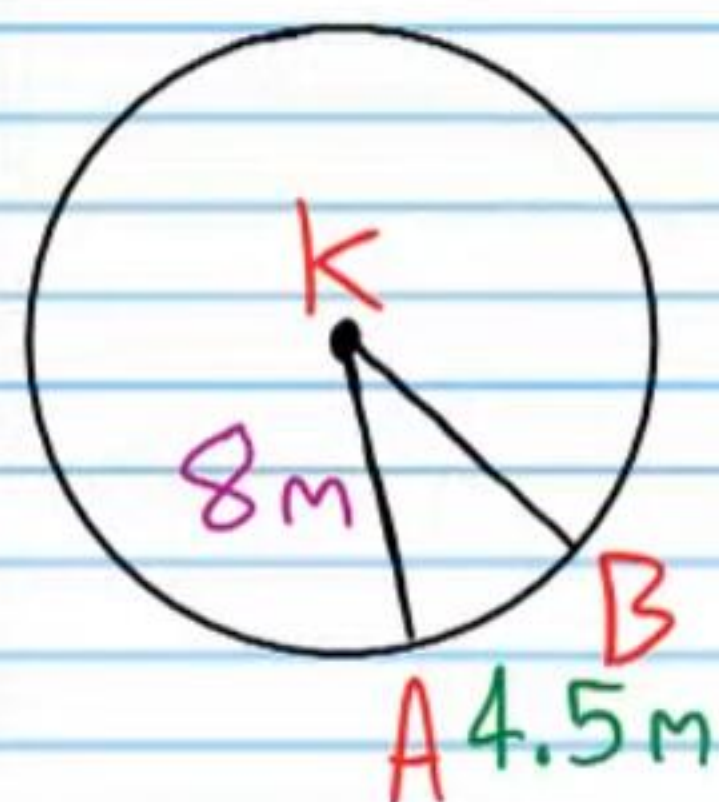


$$\mathcal{L}\widehat{LN} = \frac{\theta}{360} 2\pi r$$

$$= \frac{47}{360} 2\pi(5)$$

$$\approx \boxed{4.1 \text{ cm}}$$

⑤ $m\hat{A}B$



$$\angle AKB = \frac{\theta}{360} 2\pi r$$

$$4.5 = \frac{\theta}{360} 2\pi(8)$$

$$4.5 = \frac{\theta}{360} (16\pi)$$

$$(360) 4.5 = \frac{\theta}{360} 16\pi (360)$$

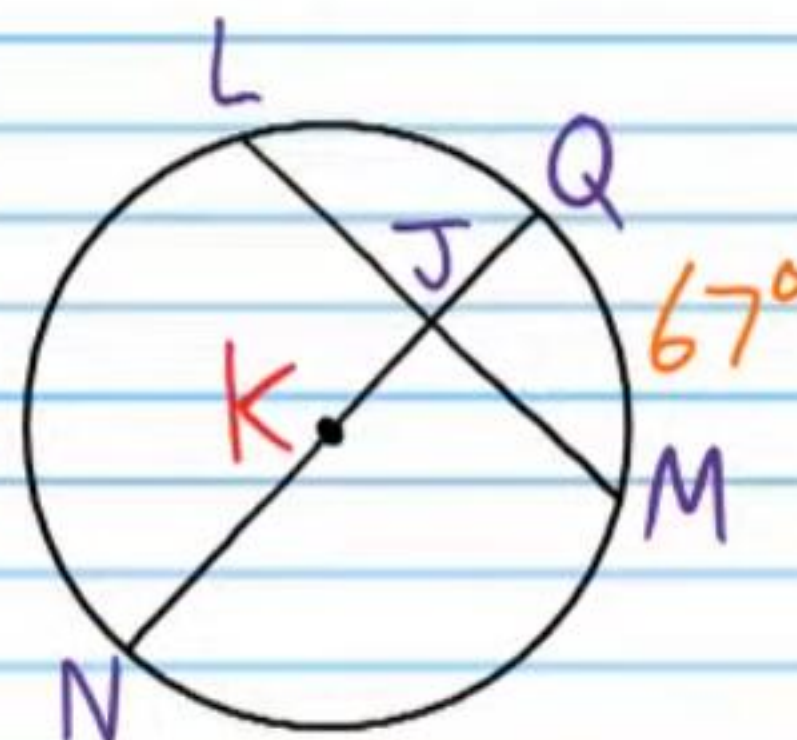
$$1620 = 16\pi\theta$$

$$\frac{1620}{16\pi} = \frac{16\pi\theta}{16\pi}$$

$$32.2^\circ \approx \theta$$

$$m\hat{A}B \approx \boxed{32.2^\circ}$$

⑥ $m\widehat{LNM}$



Not Enough Information

⑦ Find X.



$$8^2 = 6(6+x)$$

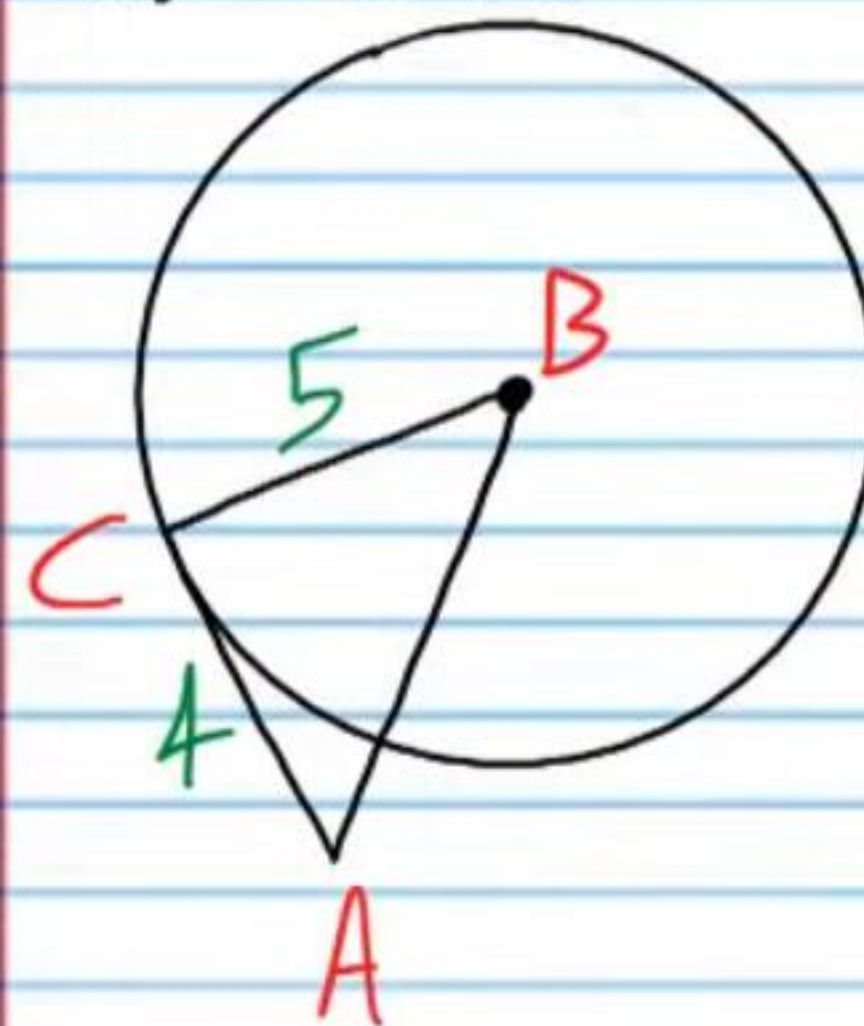
$$64 = 36 + 6x$$

$$28 = 6x$$

$$\boxed{\frac{14}{3} = x}$$

⑧ If $\overline{BC}$ is a radius, is the line containing $\overline{AC}$ tangent to $\odot B$? Show the calculations you use to determine tangency.

$$AB = 2\sqrt{10}$$



$$5^2 + 4^2 \neq (2\sqrt{10})^2$$

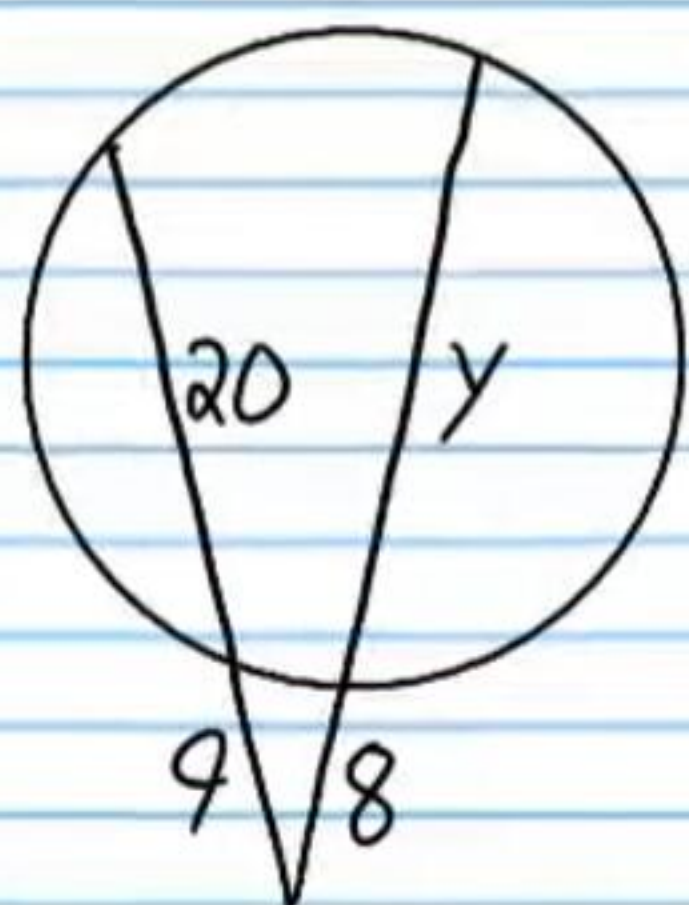
$$25 + 16 \neq 2\sqrt{10} \cdot 2\sqrt{10}$$

$$41 \neq 4(10)$$

$$41 \neq 40$$

No

⑨ Find y .



$$9(20+9) = 8(8+y)$$

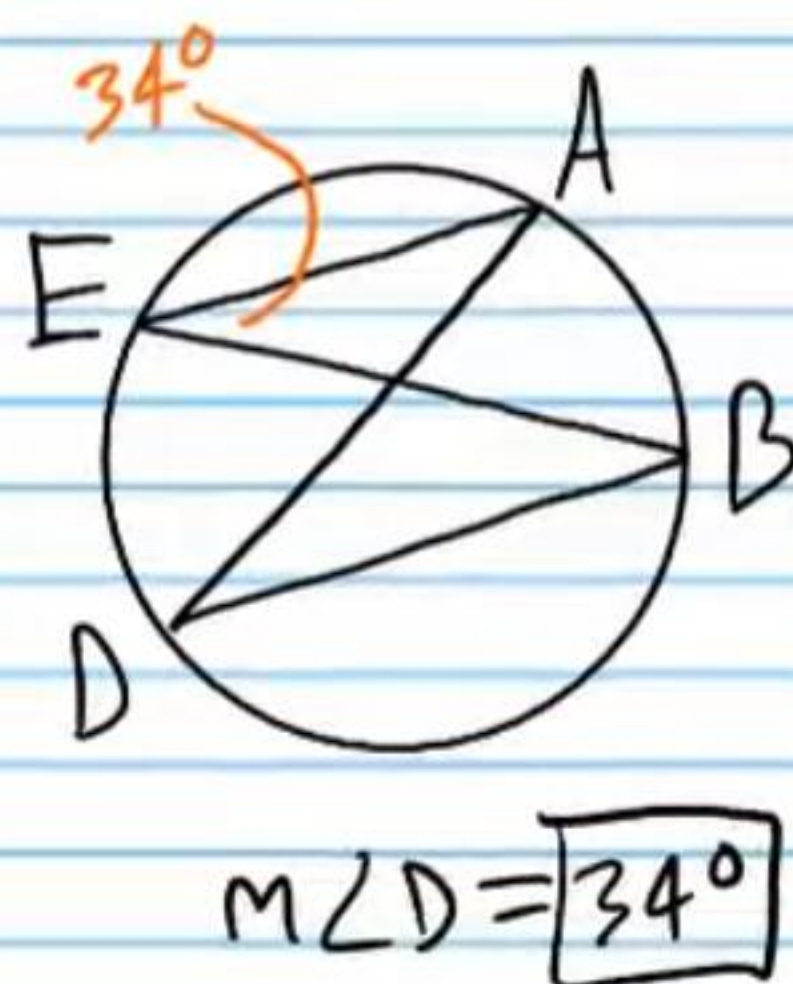
$$9(29) = 64 + 8y$$

$$261 = 64 + 8y$$

$$197 = 8y$$

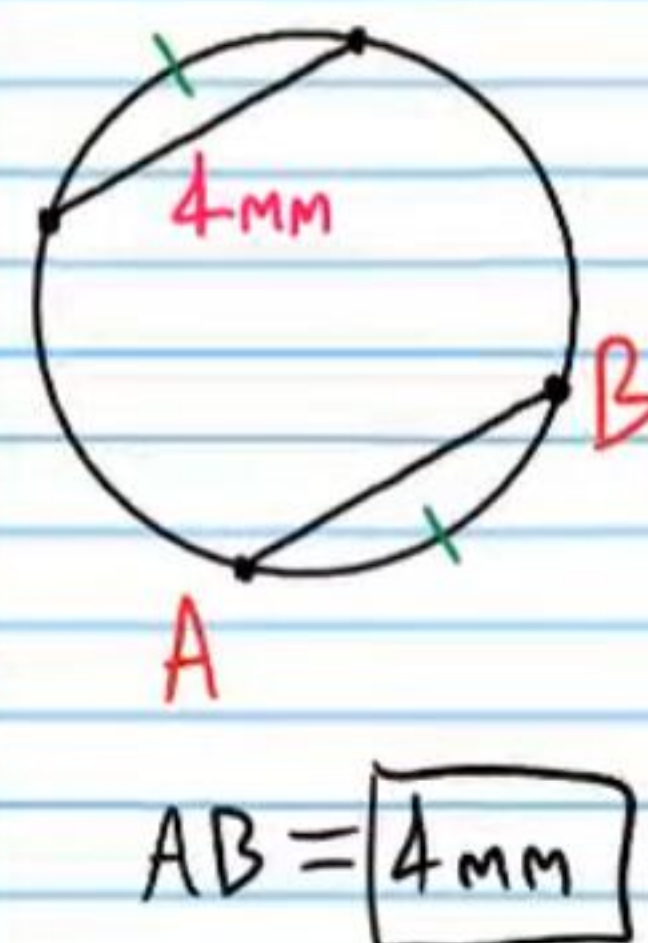
$$\boxed{24.625 = y}$$

⑩ Find $m\angle D$.



$$m\angle D = \boxed{34^\circ}$$

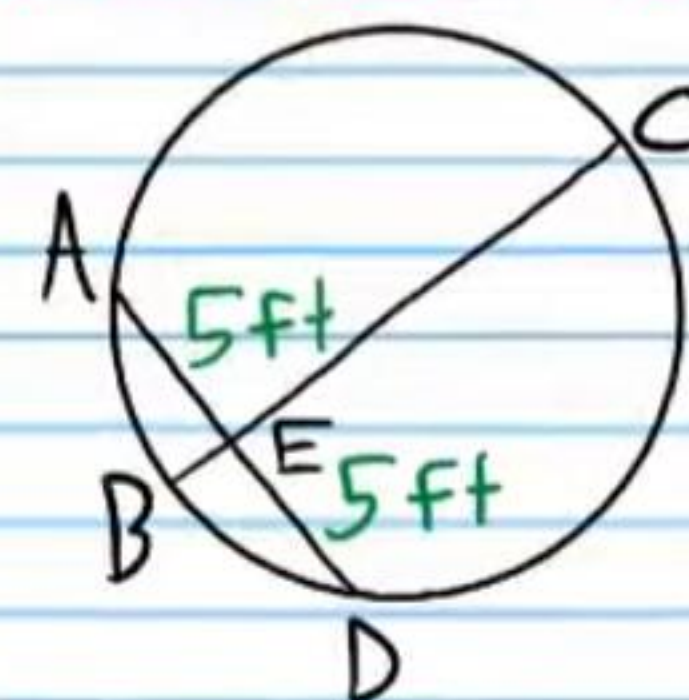
⑪ Find AB .



$$AB = \boxed{4\text{mm}}$$

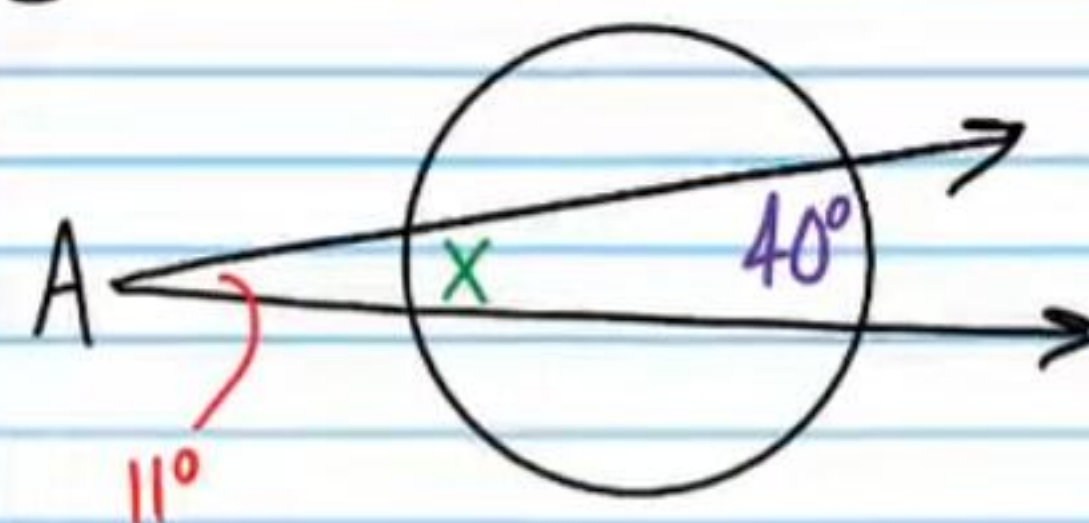
⑫ Is $\overline{BC}$ a diameter? Explain.

$$\star m\angle DEC = 90^\circ$$



Yes. $\overline{BC}$ is a $\perp$ bisector of $\overline{AD}$.

⑬ Find x .



$$11 = \frac{40 - x}{2}$$

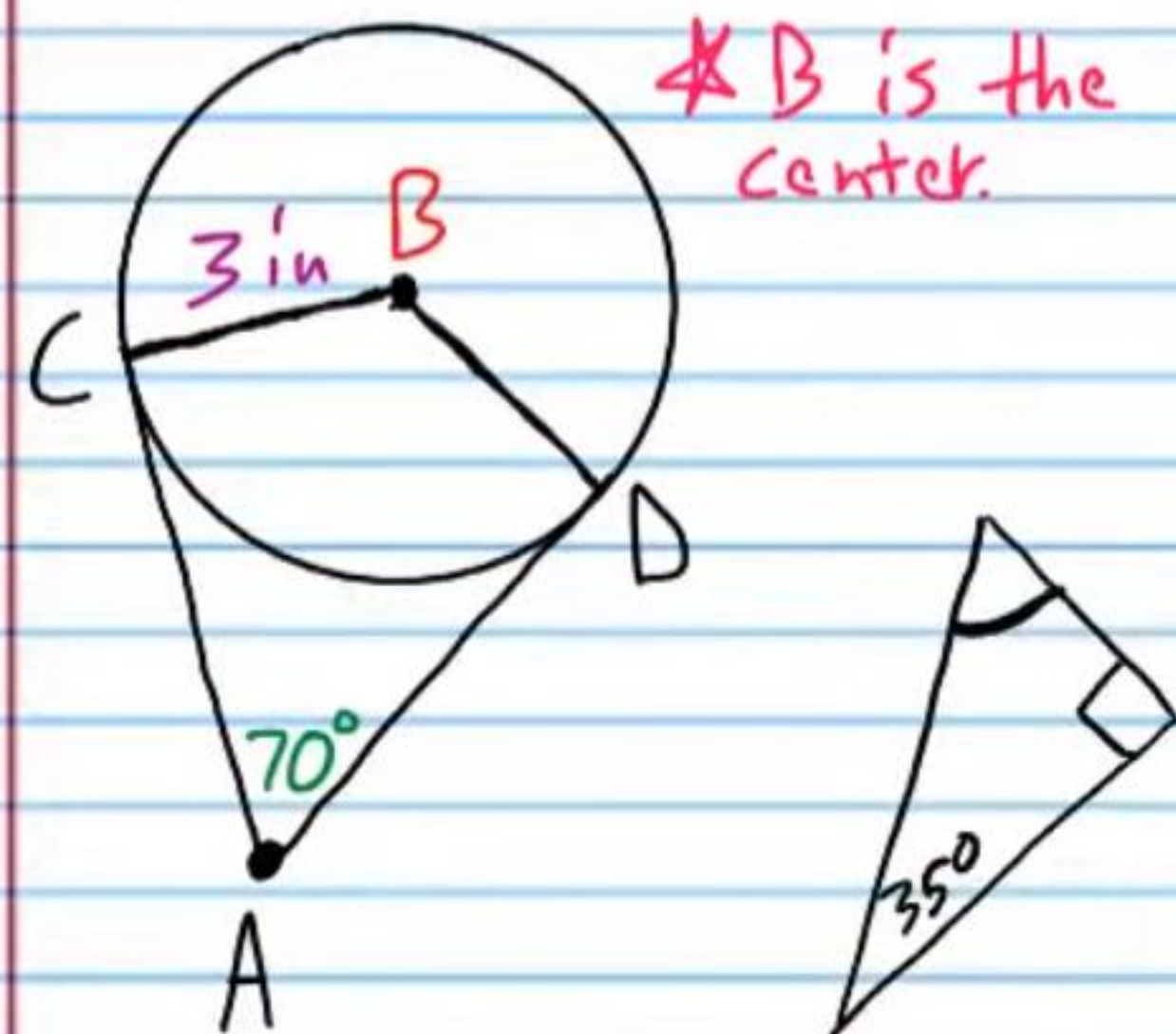
$$22 = 40 - x$$

$$-18 = -x$$

$$\boxed{18^\circ} = x$$

If C and D are points of tangency, find the following values.

⑭ i) BD & ii) $m\angle ABD$

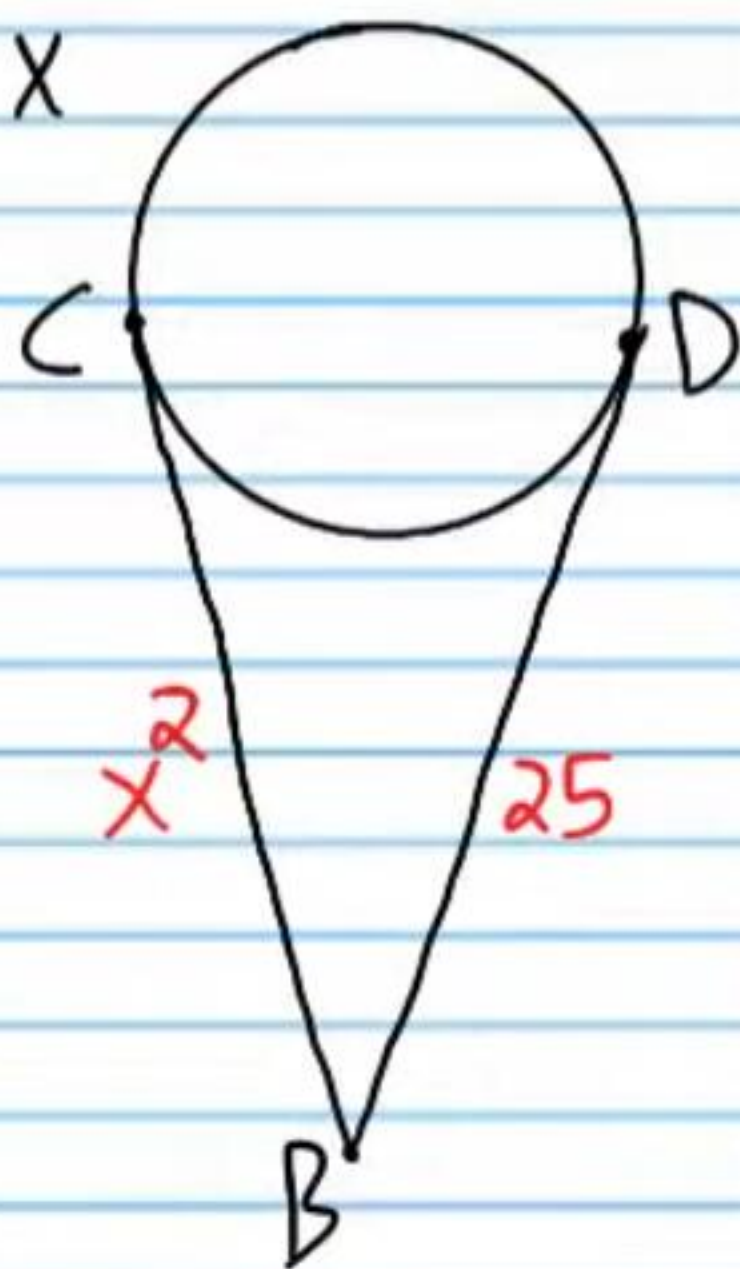


i) $BD = 3 \text{ in}$

ii) $35 + 90 + m\angle ABD = 180$

$m\angle ABD = 55^\circ$

⑮ X

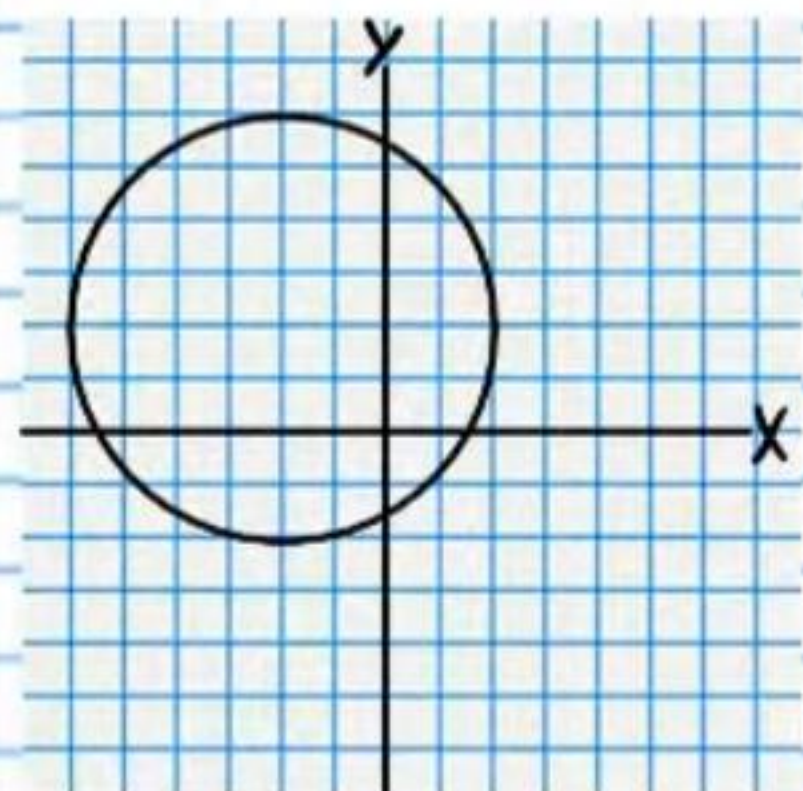


$x^2 = 25$

$\sqrt{x^2} = \pm\sqrt{25}$

$x = 5$

⑯ Write an equation of the following circle.



$(x-h)^2 + (y-k)^2 = r^2$

$C(-2, 2), r = 4$

$(x - (-2))^2 + (y - 2)^2 = 4^2$

$(x+2)^2 + (y-2)^2 = 16$

⑰ Find an equation of the circle with center $(-3, -2)$ and a point located on the circle at $(1, 4)$.

$C(-3, -2), P(1, 4)$

$(x-h)^2 + (y-k)^2 = r^2$

$(1 - (-3))^2 + (4 - (-2))^2 = r^2$

$(1+3)^2 + (4+2)^2 = r^2$

$4^2 + 6^2 = r^2$

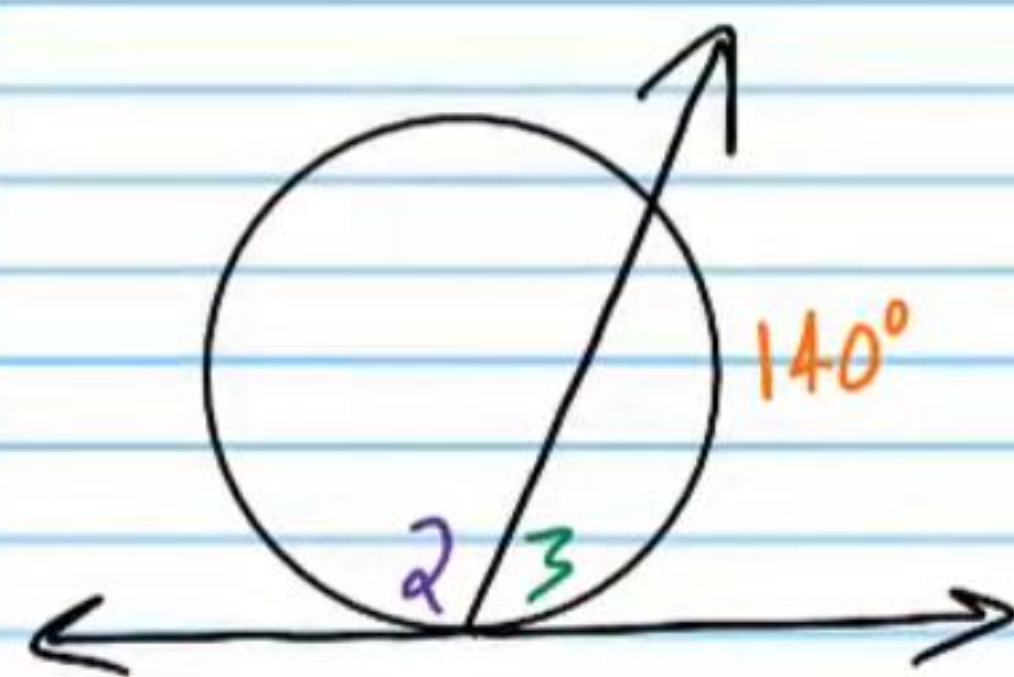
$16 + 36 = r^2$

$52 = r^2$

$(x - (-3))^2 + (y - (-2))^2 = 52$

$(x+3)^2 + (y+2)^2 = 52$

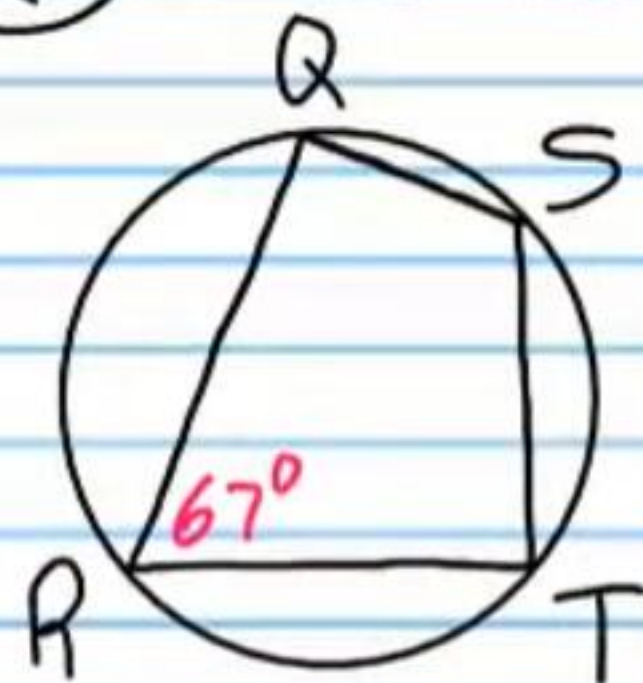
⑮ Find $m\angle 2$.



$$m\angle 3 = \frac{140}{2} = 70^\circ$$

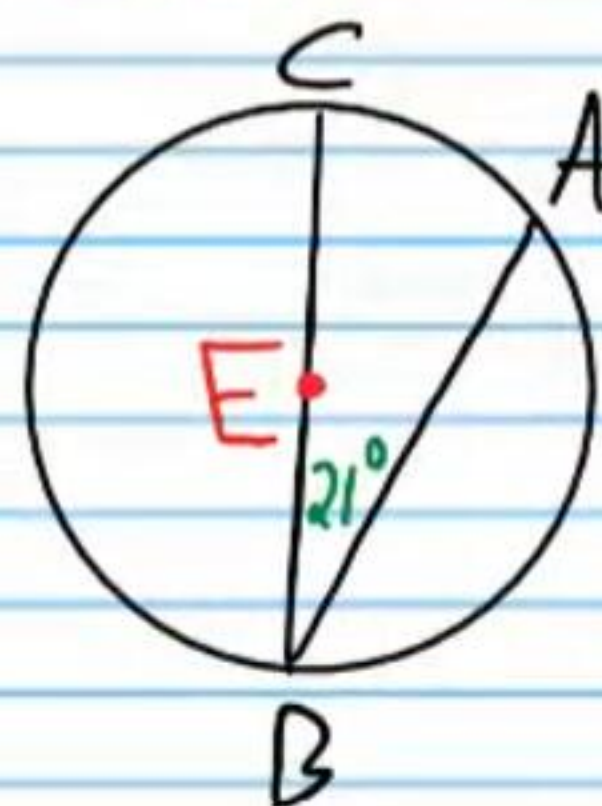
$$m\angle 2 = 180 - 70 = \boxed{110^\circ}$$

⑯ Find $m\angle S$.



$$m\angle S = 180 - 67 = \boxed{113^\circ}$$

⑰ Find $m\hat{AB}$ if E is the center of the circle below.

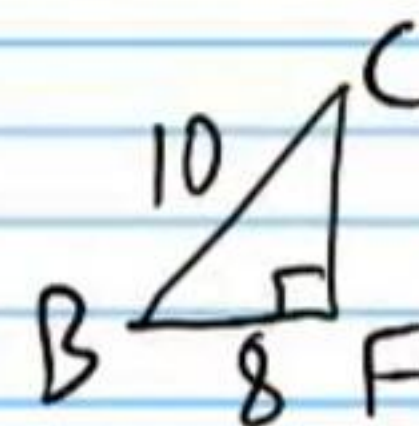
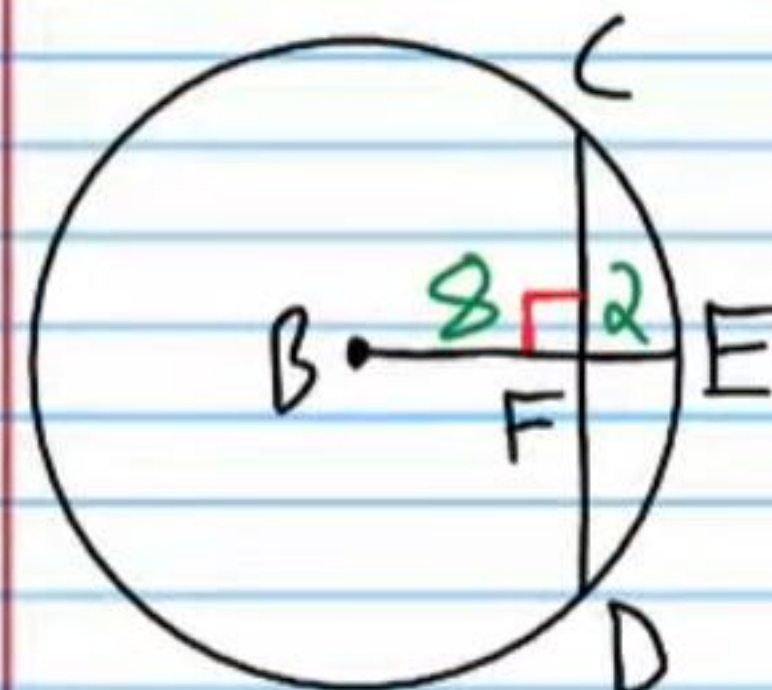


$$21 = \frac{m\hat{CA}}{2}$$

$$42 = m\hat{CA}$$

$$m\hat{AB} = 180 - 42 = \boxed{138^\circ}$$

⑱ If B is the center of the circle below, find DC.



$$10^2 = 8^2 + (CF)^2$$

$$100 = 64 + (CF)^2$$

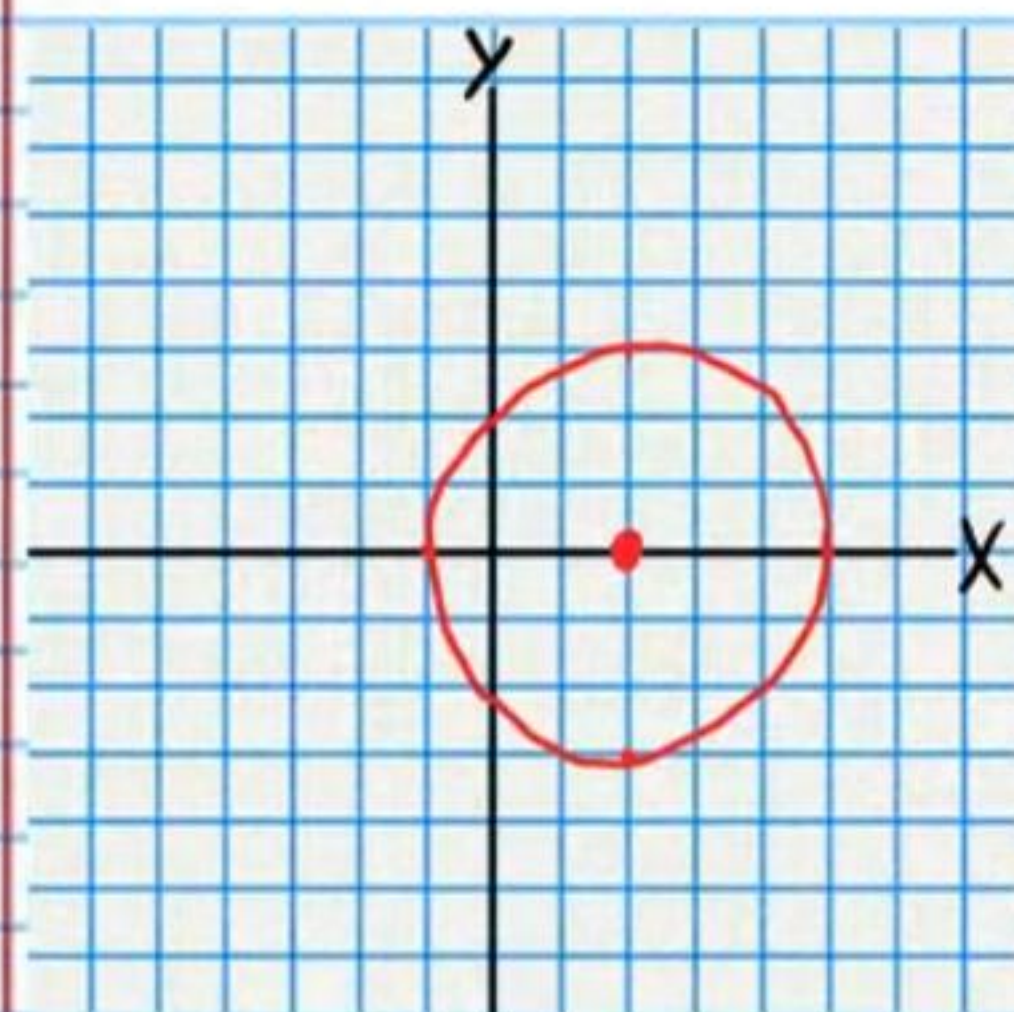
$$36 = (CF)^2$$

$$\pm\sqrt{36} = \sqrt{(CF)^2}$$

$$6 = CF$$

$$CD = 2CF = 2(6) = \boxed{12}$$

⑲ Graph the following equation: $(x-2)^2 + y^2 = 9$

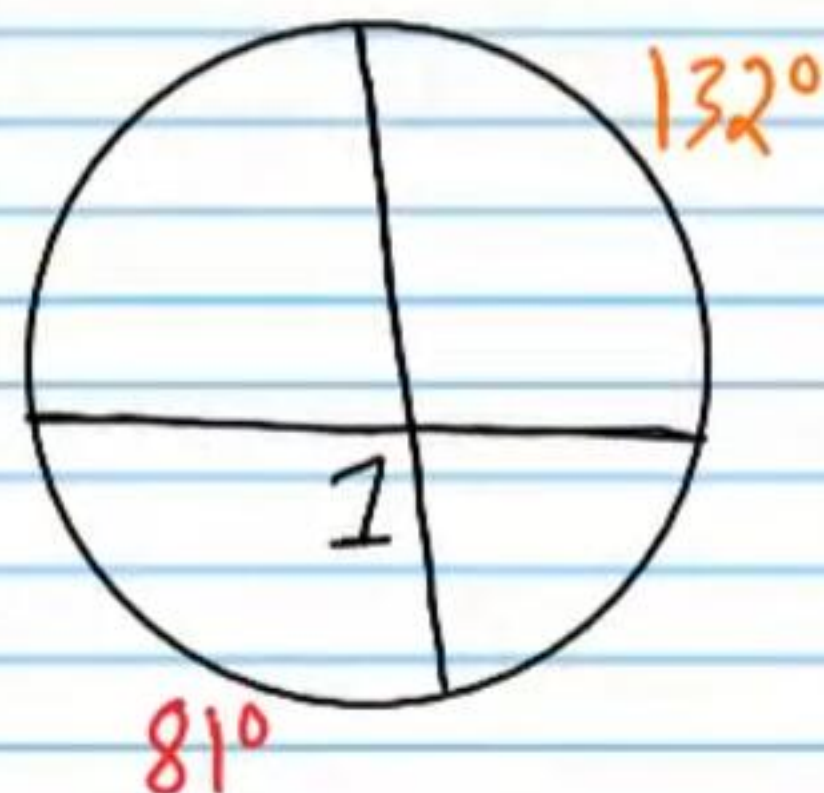


$$(x-h)^2 + (y-k)^2 = r^2$$

$$(x-2)^2 + (y-0)^2 = 3^2$$

$$C(2, 0), r = 3$$

②③ Find $m\angle 1$.

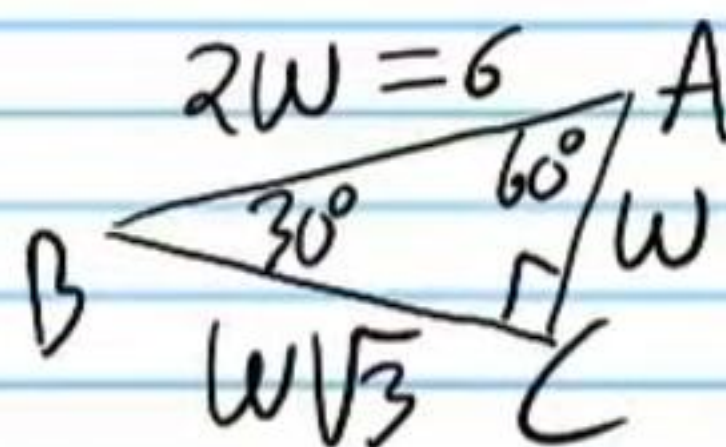
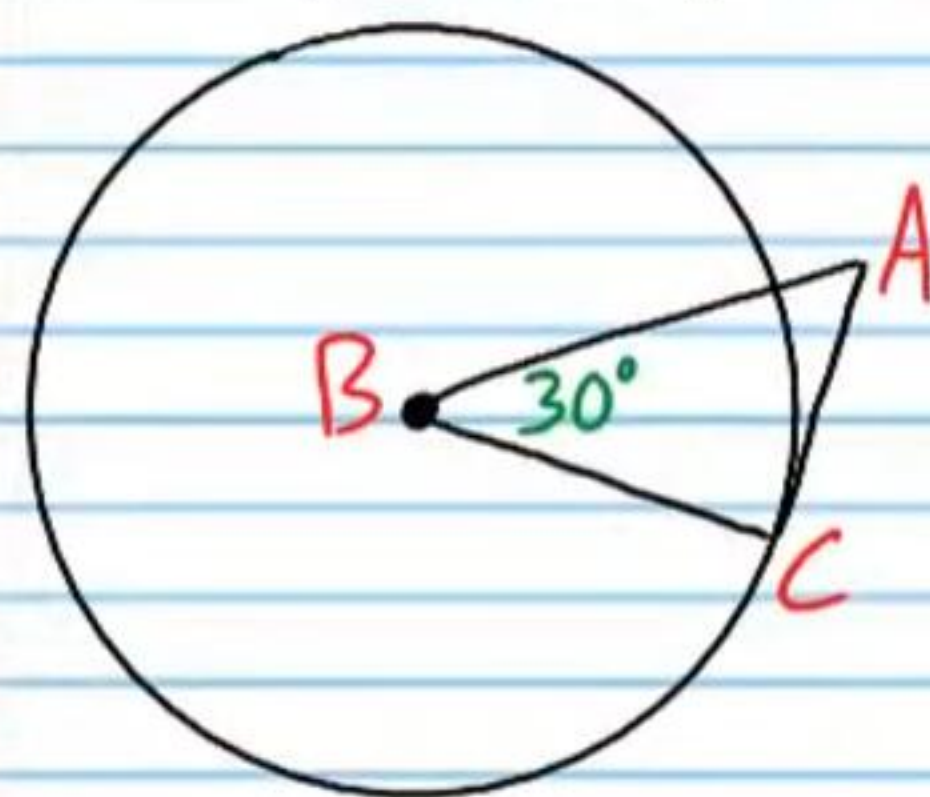


$$m\angle 1 = \frac{132 + 81}{2}$$

$$m\angle 1 = \frac{213}{2}$$

$$m\angle 1 = \boxed{106.5^\circ}$$

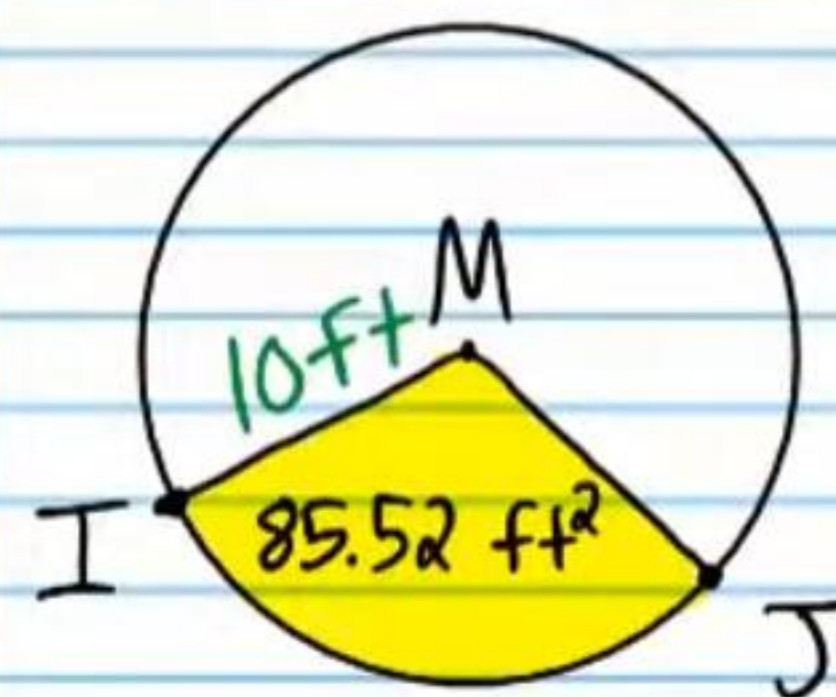
②④ If the line containing $\overline{AC}$ is tangent to $\odot B$ at C , find i) AC and ii) BC if $AB = 6$.



$$\text{i) } AC = \boxed{3}$$

$$\text{ii) } BC = \boxed{3\sqrt{3}}$$

②⑤ If M is the center of the circle below, find $m\angle IMJ$. You must use the π button on your calculator. Round to the nearest hundredth.



$$A = \frac{\theta}{360} \pi r^2$$

$$85.52 = \frac{\theta}{360} \pi (10)^2$$

$$85.52 = \frac{\theta}{360} \pi (100)$$

$$(360) 85.52 = \frac{\theta}{360} \pi (100) (360)$$

$$30,787.2 = 100\pi\theta$$

$$\frac{30,787.2}{100\pi} = \frac{100\pi\theta}{100\pi}$$

$$98^\circ \approx \theta$$

$$\boxed{98^\circ} \approx m\angle IMJ$$